UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 005 · Particular integral Type 3 (X = x^m) · compiled 16 October 2026

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1 Short notes

1 · The Type 3 rule: $X = x^m$

Type 3: $X = x^m$. PI $= \frac{1}{f(D)}x^m = [f(D)]^{-1}x^m$. Take the lowest-degree term out, $f(D) = a_0[1 \pm \varphi(D)]$; write $\frac{1}{a_0}[1 \pm \varphi(D)]^{-1}$; expand by the binomial theorem up to D^m (since $D^{m+1}x^m = 0$); operate term by term with $D = \frac{d}{dx}$. Examples: $\frac{5x-2}{D^2-1} = 2 - 5x$; $\frac{x^2+7x+9}{D^2+5D+4} = \frac{1}{4}(x^2 + \frac{9}{2}x + \frac{23}{8})$; CW P15: $\frac{2x^2+6x}{D^2-2D-3} = -\frac{18x^2+30x-8}{27}$.

2 · The binomial expansions and where to stop

$(1+z)^{-1} = 1 - z + z^2 - z^3$, $(1-z)^{-1} = 1 + z + z^2 + z^3$, $(1+z)^{-2} = 1 - 2z + 3z^2 - 4z^3$, $(1-z)^{-2} = 1 + 2z + 3z^2 + 4z^3$. In squares and cubes keep only products with D-power $\le m$: for $m = 3$, $(D^2 + 3D)^2 \to 9D^2 + 6D^3$, $(D^2 + 3D)^3 \to 27D^3$. $\frac{x^3+x^2}{D^2+3D+2} = \frac{1}{2}(x^3 - \frac{7x^2}{2} + \frac{15x}{2} - \frac{31}{4})$; $\frac{x^2+2x-1}{2D^2+2D+3} = \frac{1}{3}(x^2 + \frac{2x}{3} - \frac{25}{9})$; CW P16: $\frac{X}{D^3-1} = -(1 + D^3)X$.

3 · When D is a factor of $f(D)$

No constant term in $f(D)$: take D^k out, $\frac{1}{D^k}[\psi(D)]^{-1}x^m$; expand, operate, then integrate k times without constants. The PI rises k degrees. $\frac{x^2+2x+4}{D^2+D} = \frac{x^3}{3} + 4x$; $\frac{6x}{D^3+D^2} = x^3 - 3x^2$. A constant: $\frac{k}{f(D)} = \frac{k}{f(0)}$.

4 · Mixed right-hand sides: Types 1, 2 and 3 together

Mixed X: CF first (it shows which terms fail), then PI_1+PI_2+PI_3, each with its own rule: $e^{ax} \to D = a$; $a^{-x} = e^{-x\log a} \to D = -\log a$; $\sin, \cos \to D^2 = -a^2$; polynomial → series. ESA: $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$: PI $= \frac{e^{4x}}{2} + \frac{\sin x + \cos x}{10} + \frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$. $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$: PI $= -xe^{2x} - \frac{x}{6} - \frac{11}{36}$.

2 Keyword index

18 terms; the number in brackets is the topic.

$(1+z)^{-1}$ (2). $1 - z + z^2 - z^3 + \dots$

$(1+z)^{-2}$ (2). $1 - 2z + 3z^2 - 4z^3 + \dots$

$(1-z)^{-1}$ (2). $1 + z + z^2 + z^3 + \dots$

$(1-z)^{-2}$ (2). $1 + 2z + 3z^2 + 4z^3 + \dots$

$[f(D)]^{-1}$ (1). The inverse operator, written as a series.

$\varphi(D)$ (1). $f(D) = a_0[1 \pm \varphi(D)]$.

a^{-x} (4). $e^{-x \log a}$: Type 1 with $D \to -\log a$.

$D^{m+1}x^m = 0$ (1). Why the series stops.

Ascending powers of D (2). $1 + c_1 D + c_2 D^2 + \dots$

Binomial theorem (2). Expands $[1 \pm \varphi(D)]^{-1}$.

Degree rise (3). PI degree $m + k$ with a factor D^k.

Factor D (3). No constant term; $\frac{1}{D}$ integrates.

Lowest-degree term (1). Taken out of $f(D)$ first.

Mixed right-hand side (4). Polynomial plus e^{ax}, a^x, $\sin$, $\cos$.

Operate term by term (1). D differentiates, D^2 twice, ...

Superposition (4). PI = PI_1+PI_2+PI_3.

Truncation (2). Drop products with D-power $> m$.

Type 3 (1). $X = x^m$ or any polynomial.

3 Rapid recall

1	Type 3 PI?	$\frac{1}{f(D)}x^m = [f(D)]^{-1}x^m$, expanded in ascending powers of D.
2	First step of the Type 3 rule?	Take the lowest-degree term out: $f(D) = a_0[1 \pm \varphi(D)]$.
3	How far to expand for x^m?	Up to D^m; $D^{m+1}x^m = 0$.
4	$(1+z)^{-1}$?	$1 - z + z^2 - z^3 + \dots$
5	$(1-z)^{-2}$?	$1 + 2z + 3z^2 + 4z^3 + \dots$
6	$\frac{5x-2}{D^2-1}$?	$-(1 + D^2 + \dots)(5x - 2) = 2 - 5x$.
7	$\frac{x^2+7x+9}{D^2+5D+4}$?	$\frac{1}{4}(x^2 + \frac{9}{2}x + \frac{23}{8})$.
8	$\frac{x^3+x^2}{D^2+3D+2}$?	$\frac{1}{2}(x^3 - \frac{7x^2}{2} + \frac{15x}{2} - \frac{31}{4})$.
9	$\frac{x^2+2x-1}{2D^2+2D+3}$?	$\frac{1}{3}(x^2 + \frac{2x}{3} - \frac{25}{9})$.
10	In $(D^2+3D)^2$ for $m=2$, what is kept?	Only $9D^2$.
11	Roots of $2m^2 + 2m + 3 = 0$?	$\frac{-1 \pm \sqrt{5}i}{2}$ (slide 005·13 prints $\frac{-2 \pm \sqrt{5}i}{2}$).
12	$\frac{x^5+3x^4-2x^3}{D^3-1}$ (CW P16)?	$-x^5 - 3x^4 + 2x^3 - 60x^2 - 72x + 12$.
13	$\frac{x^2+2x+4}{D^2+D}$?	$\frac{1}{D}(x^2+4) = \frac{x^3}{3} + 4x$.
14	$\frac{8x^2}{(D-2)^2}$?	$2(1 - \frac{D}{2})^{-2}x^2 = 2x^2 + 4x + 3$.
15	$\frac{x}{D^3-6D^2+11D-6}$?	$-\frac{x}{6} - \frac{11}{36}$.
16	$\frac{2^{-x}}{D^2+4}$?	$\frac{2^{-x}}{(\log 2)^2+4}$.
17	PI of $(D^2-5D+6)y = e^{4x} + \sin x + x^2$?	$\frac{e^{4x}}{2} + \frac{\sin x + \cos x}{10} + \frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$.
18	PI of $(D^2+4)y = x^2 + \cos 2x + 2^{-x}$?	$\frac{x^2}{4} - \frac{1}{8} + \frac{x \sin 2x}{4} + \frac{2^{-x}}{(\log 2)^2+4}$.

Mnemonic (study aid). *"Lowest out, bracket up, stop at m, differentiate."* The four steps of slide 005·4 (study aid).

4 Last 10 minutes before the exam

Top facts

1. Type 3: $\frac{1}{f(D)}x^m = \frac{1}{a_0}[1 \pm \varphi(D)]^{-1}x^m$: lowest-degree term out, expand, operate.
2. Expand only up to D^m (m = degree of X): $D^{m+1}x^m = 0$. In squares and cubes keep only products with D-power $\le m$.
3. $(1+z)^{-1} = 1 - z + z^2 - \ldots$, $(1+z)^{-2} = 1 - 2z + 3z^2 - \ldots$; for $1 - z$ all signs are plus.
4. No constant term in $f(D)$: take D^k out, integrate k times at the end (no constants).
5. Mixed X: CF first, then PI_1+PI_2+PI_3; $a^{-x} = e^{-x\log a}$; $\sin^2\frac{x}{2} = \frac{1-\cos x}{2}$.
6. ESA ×4: $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$; $(D^2 + 4)y = x^2 + \cos 2x + 2^{-x}$; $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$; $(D^2 - 1)^2 y = \sin^2\frac{x}{2} + 2^{-x} + x$.
7. Slide slips: 005·7 should read $-(1 + D^2 + D^4 + \ldots)$; 005·13 roots are $\frac{-1\pm\sqrt{5}i}{2}$ and the CF needs c_2.

Top formulas

E1.2 · Take the lowest-degree term out

ESA ★★★

$$\frac{1}{f(D)}x^m = \frac{1}{a_0\,[1 \pm \varphi(D)]}x^m = \frac{1}{a_0}\big[1 \pm \varphi(D)\big]^{-1}x^m$$

GIVES	A bracket ready for the binomial series.
APPLIES	a_0 = the lowest-degree term (usually the constant).
NOTE	Take out the constant with its sign: $D^2 - 1 = -(1 - D^2)$, $D^2 + 5D + 4 = 4\big[1 + \frac{D^2+5D}{4}\big]$.

S005·4 · S005·7 · S005·8 · T005 00:01:08 · T005 00:09:16

E1.3 · Powers of D on x^m

ESA ★★★

$$D^k x^m = \frac{m!}{(m-k)!}\,x^{m-k}\ (k \le m), \qquad D^k x^m = 0\ (k > m)$$

GIVES	Why the series stops at D^m; $D^m x^m = m!$ is a constant.
APPLIES	Operating term by term, $D = \frac{d}{dx}$.
NOTE	Example: $D(x^2 + 7x + 9) = 2x + 7$, $D^2(x^2 + 7x + 9) = 2$, $D^3(\ldots) = 0$.

S005·4 · S005·10 · T005 00:01:51 · T006a 00:01:29 · T006a 00:02:06

E2.1 · First-power series

ESA ★★★

$$(1+z)^{-1} = 1 - z + z^2 - z^3 + \dots$$
$$(1-z)^{-1} = 1 + z + z^2 + z^3 + \dots$$

GIVES $[1 \pm \varphi(D)]^{-1}$ with $z = \varphi(D)$.

APPLIES After E1.2.

NOTE Signs alternate for $(1+z)^{-1}$; all plus for $(1-z)^{-1}$ (replace z by $-z$).

S005·5 · T005 00:02:45 · T005 00:03:12 · T006a 00:04:36

E2.2 · Second-power series

ESA ★★

$$(1+z)^{-2} = 1 - 2z + 3z^2 - 4z^3 + \dots$$
$$(1-z)^{-2} = 1 + 2z + 3z^2 + 4z^3 + \dots$$

GIVES $[1 \pm \varphi(D)]^{-2}$ when $f(D)$ is a square, e.g. $(D-2)^2$, $(D^2-1)^2$.

APPLIES $f(D) = a_0^2[1 \pm \varphi(D)]^2$.

NOTE "If we remember the first and the third, it is sufficient."

S005·5 · T005 00:03:46 · T005 00:04:04 · CW U2 P21 · CW U2 P22

E2.3 · Keep only powers up to D^m

ESA ★★★

$$(D^2+3D)^2 = D^4 + 6D^3 + 9D^2 \;\to\; 6D^3 + 9D^2$$
$$(D^2+3D)^3 = D^6 + \dots + 27D^3 \;\to\; 27D^3 \qquad (m=3)$$

GIVES The terms of the square and cube that survive.

APPLIES $\varphi(D)$ has two or more terms.

NOTE Lecture: "we don't want a^2, we want b^2, ... $2ab$ that is $6D^3$".

S005·12 · T006a 00:06:05 · T006a 00:06:22 · T006a 00:06:43 · T005 00:11:24

E4.1 · Superposition

ESA ★★★

$$\frac{1}{f(D)}(X_1 + X_2 + X_3) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2 + \frac{1}{f(D)}X_3 = \mathrm{PI}_1 + \mathrm{PI}_2 + \mathrm{PI}_3$$

GIVES The PI of a mixed right-hand side.

APPLIES X is a sum of terms of different types.

NOTE Each term chooses its own rule and its own failure test.

ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c · CW U2 P21